

Activity 3.6 Practical Skills in Biology: Marking scheme

Name

Tasks	Outcomes tested	Marks
1. Using a microscope	Slide in focus on low power (1), Focused on high power using fine focus only (1), Correct magnification calculated (2).	/4
2. Making a wet mount	Coverslip central on slide (1/2), Small specimen underneath (1/2), Flat specimen (1), No air bubbles (1), Correct amount of water (1), Well stained (1).	/5
3. Table of results	Appropriate heading that describes the data (1), Independent variable left column & dependent variable in right column (1), appropriate headings at top of columns (1), units in column heading (1).	/4
4. Biological drawing	a) Biological drawing Good size drawing (1/2), Drawn freehand (1/2), No shading (1/2), Ruled label lines (1/2), Horizontal label lines (1/2), Pencil used (1/2), Labelled cell wall, cell membrane, chloroplast and cytoplasm (1/2 each)	/5
	b) Plant or animal cell? Identified as plant cell (1), Reason: one of - cell wall, chloroplasts, regular shape (1)	/2
	c) Plant or animal cell? Identified as animal cell (1), Reason: No cell wall, irregular shape, appendages, not green (no chloroplasts) (1)	/2
	d) Measurement under a microscope Width of organism between 380 - 460 μm (2) (1 mark off if no units given)	/2
5. Scale drawings	a) Calculating size Size of coral trout 46 - 52 cm (1), Working shown (1)	/2
	b) Scale drawing Appropriate heading (1/2), Pencil used (1/2), Correct sized drawing (1/2), Scale shown (1/2), Drawn freehand (1/2), No shading (1/2), Ruled label lines (1/2), Horizontal label lines (1/2).	/4
Are you a good research scientist? Total mark =		/30

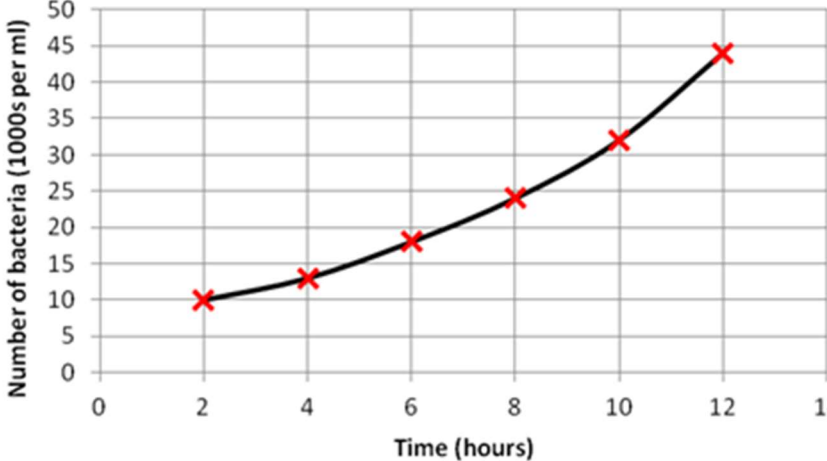
Activity 5.9 - Topic Test Marking Scheme

Question	Criteria											Marks Allotted	
1-10	Multiple Choice section:											/10	
		1	2	3	4	5	6	7	8	9	10		
		D	B	C	B	A	C	A	D	A	C		
11	A: chloroplast B: cell wall C: vacuole D: mitochondrion E: cytoplasm F: nucleus 1 mark each correct answer											/6	
12	a) They become adapted to do a particular job											/1	
	b) e.g. Muscle cells are long and thin and can contract to move limbs. Secretory cells produce chemicals e.g. enzymes for digestion. (many answers possible)											/1	
13	3 x 1 mark for correct statements with named examples. - A tissue is a group of similar cells doing a particular job e.g. muscle cells make up muscle tissue - An organ is a body part made of several tissues that perform a particular function for the body. e.g. the heart is made up of muscle tissue, nerve tissue and blood. - A system consists of a group of organs working together to perform a body function. e.g. the circulatory system consists of the heart, arteries, veins and capillaries working together to carry cell requirements and waste around the body.											/3	
	1 mark for well-explained and linked examples											/1	
14	½ mark each correct statement (or similar)											/4	
	Part of skeleton	Kangaroo					Dolphin						
	Forelimb	Short strong arms with paws/claws for grasping.					Modified as flippers for swimming						
	Hind limb	Long and strong for hopping.					None present (vestigial bone) as not needed in water						
	Tail	Long for balance.					Flat for swimming						
15	Backbone											/1	
	Strong and curved to hold up body (interlocking vertebrae)												
15	a) 40mm = 40,000 μm Alga B is 40,000/10 = 4,000 times bigger											/1	
	b) If a cell becomes too large its SA : V ratio becomes too small to allow efficient diffusion of materials in and out of the cell.											/1	
	c) It divides (mitosis) to give 2 smaller cells.											/1	
	d) The many functions needed to keep the organism alive can be shared between different kinds of specialised cells.											/1	

Topic Test Marking Scheme (continued)

16	1 mark each: a) D b) C c) A d) names an adaptation. e.g. coloured petals and nectar lures an insect into flower - Gives a reason for adaptation: e.g. makes sure that insect touches stigma to transfer pollen.	/3 /1 /1
17	Example answer: A van Leeuwenhoek (6 marks). a) microbiologist b) first person to see unicellular organisms c) he shared his findings with the Royal Society spreading the interest in cell biology OR he built many microscopes so that there were improvements in microscopes and the detail that could be seen in cell structure e.g. nucleus discovered, cell theory etc. d) Led to the discovery of chromosomes in the nucleus and so has had a big influence on our increasing understanding of inheritance.	/1 /1 /2 /2
18	1 mark each a) lungs b) circulatory system c) respiration d) carbon dioxide e) oxygen f) provide energy for metabolism.	/6
19	a) it is essential for life b) two of (1 mark each): reproduction, react to stimulus (light), organised (made of cells), growth c) may absorb light energy (photosynthesis) OR may consume other organisms d) discussion with three points e.g. shows some characteristics of living things but cannot observe all; seem to be made of cells, grow, reproduce and react to changes; Green so may contain chlorophyll/photosynthesis. Summarising statement - cannot be certain they are alive but good evidence that this is the case. e) to confirm they could (two reasonable suggestions needed) e.g. • test for genetic material /DNA • test if they can survive in the dark or what nutrients they need to survive • study the organism under an electron microscope to verify if made of cells.	/1 /2 /1 /4 /2
20	a) both have a good supply of blood capillaries to absorb oxygen OR both have a large surface area to maximise gas exchange b) gills are on the outside of the fish's body, lungs are internal c) there is not enough oxygen dissolved in water to supply the lungs efficiently AND water would be unable to be expelled, so would drown.	/1 /1 /1 /1
21	a) An endoscope is a camera on the end of a tube, which is passed via the mouth into the stomach. It enables viewing of the stomach lining without invasive surgery.	/2

Topic Test Marking Scheme (continued)

21	b) Two of (1 mark each): - He determined that he did not have <i>Helicobacter</i> bacteria or ulcers in his stomach before the experiment. - He confirmed that his disease symptoms occurred with the presence of <i>Helicobacter</i> in his stomach during the test. - He showed that the symptoms disappeared when the bacteria was killed by antibiotics. c) No. He did not develop ulcers (only gastritis) AND he needed to do the experiment on more people to check the results (replicate). d) (i) <u>Graph</u>: Graph to show bacterial growth over time <table><caption>Data points from the bacterial growth graph</caption><tr><th>Time (hours)</th><th>Number of bacteria (1000s per ml)</th></tr><tr><td>2</td><td>10</td></tr><tr><td>4</td><td>13</td></tr><tr><td>6</td><td>18</td></tr><tr><td>8</td><td>24</td></tr><tr><td>10</td><td>32</td></tr><tr><td>12</td><td>45</td></tr></table> 1 mark - points plotted correctly $\frac{1}{2} \times 2$ - scales spread on axes $\frac{1}{2} \times 2$ - correct scales $\frac{1}{2} \times 2$ - axes labelled correctly including units 1 - line of best fit - 1 - if axes wrong way around -1 - if column graph (ii) The number of bacteria increases more quickly with time. (iii) They will be genetically identical to the parent cells. (iv) No. The solution containing the bacteria would run out of nutrients eventually OR become overcrowded, so growth would slow down.	Time (hours)	Number of bacteria (1000s per ml)	2	10	4	13	6	18	8	24	10	32	12	45	/2 /2 /5 /1 /1
Time (hours)	Number of bacteria (1000s per ml)															
2	10															
4	13															
6	18															
8	24															
10	32															
12	45															
TOTAL		/75														